

HIGH GRADE

Nytro Gemini X

High performance insulating oil

An inhibited high-grade oil, Nytro Gemini X conforms to IEC 60296 Edition 4.0 – including the fulfilment of specific requirements for special applications. Developed and formulated to deliver strong resistance to oil degradation, Nytro Gemini X provides excellent oxidation stability for a longer transformer life with less maintenance.

Designed for heavy duty

This product has been specially developed for use in oil-filled electrical equipment – including power and distribution transformers, rectifiers, circuit breakers and switchgears.

Performance and benefits

Very good heat transfer. Thanks to low viscosity and viscosity index, this high grade offers extremely good heat transfer characteristics, ensuring heat is efficiently removed from core and windings.

Excellent oxidation stability. Developed and formulated to deliver superb resistance to oil degradation, this grade provides excellent oxidation stability for enhanced transformer life and minimum maintenance.

Very good low temperature properties. Naphthenic characteristics allow the transformer to start at the lowest possible temperature – without using pour point depressants.

High dielectric strength. This insulating oil both meets and exceeds the toughest demands on dielectric strength – when stored and handled correctly.

Product description

Nytro Gemini X fulfils the requirements for IEC 60296 Edition 4.0 fully inhibited oil. Nynas classify this product as a high grade.

Nytro Gemini X is rigorously analysed and passes the following corrosion tests:

- ASTM D1275 method B
- IEC 62535
- DIN 51353

In accordance with IEC 60296 Edition 4.0, all additives are declared.

There's more to us than this

We're delighted you chose one of our transformer oils. If you have any questions about other products and services, get in touch with your local Nynas contact. Besides top quality oils, we offer a wide range of services, including rapid delivery worldwide, sample analysis, training, seminars and much more. All you have to do is ask. Find out more at www.nynas.com



Nytro Gemini X

PROPERTY	UNIT	TEST METHOD	SPECIFICATION LIMITS		TYPICAL DATA
			MIN	MAX	
1 - Function					
Viscosity, 40 °C	mm ² /s	ISO 3104		12.0	9.0
Viscosity, -30 °C	mm ² /s	ISO 3104		1800	800
Pour point	°C	ISO 3016		-40	-51
Water content	mg/kg	IEC 60814		30	<20
Breakdown voltage					
- Before treatment	kV	IEC 60156	30		40-60
- After treatment	kV		70		>70
Density, 20 °C	kg/dm ³	ISO 12185		0.895	0.870
DDF at 90 °C		IEC 60247		0.005	<0.001
2 - Refining/stability					
Appearance		IEC 60296	Clear, free from sediment		complies
Acidity	mg KOH/g	IEC 62021		0.01	<0.01
Interfacial tension	mN/m	EN 14210	40		50
Total sulphur content	%	ISO 14596		0.15	0.01
Corrosive sulphur		DIN 51353	non-corrosive		non-corrosive
Potentially corrosive sulphur		IEC 62535	non-corrosive		non-corrosive
Corrosive sulphur		ASTM D 1275 B	non-corrosive		non-corrosive
DBDS	mg/kg	IEC 62697-1		not detectable	not detectable
Antioxidant	wt %	IEC 60666		0.4	0.38
Metal passivator additives	mg/kg	IEC 60666		not detectable	not detectable
2-Furfural and related compounds content	mg/kg	IEC 61198		0.05	<0.05
Aromatic content	%	IEC 60590			3
3 - Performance					
Oxidation stability at 120 °C,500 h		IEC 61125 C			
Total acidity	mg KOH/g			0.3	0.06
Sludge	wt %			0.05	<0.02
DDF at 90 °C				0.050	0.01
4 - Health, safety and environment (HSE)					
Flash point, PM	°C	ISO 2719	135		152
PCA	wt %	IP 346		3	<3
PCB		IEC 61619	not detectable		not detectable

Nytro Gemini X is an inhibited insulating oil with extremely good electrical and excellent ageing properties meeting IEC 60296 Ed.4 (2012), special applications.

Severely Hydrotreated Insulating Oil
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